Tab 1

User Documentation

SI 495 - Information Analysis Capstone II

Group 3

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Ann Arbor Hands-On Museum

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Ann Arbor Hands-On Museum Dashboard Project
User Documentation

1. Project Overview

This project was completed as part of the **SI 485 Capstone course** at the University of Michigan in collaboration with the Ann Arbor Hands-On Museum.

The goal of the project is to **build interactive dashboards** that help the museum better understand attendance, membership, and donor behavior. These insights support decision-making, improve engagement, and help the museum reach more communities.

2. Objectives

- Analyze museum data to identify trends and patterns
- Build 3 dashboards.
- Support data-driven decision-making

3. Deliverables:

Dashboard Overview

To develop the three dashboards for this project we used the Attendance dataset and added year-over-year comparison of 52 weeks attendance.

- The first dashboard is t**he attendance dashboard,** which include visualizations that track attendance across the museum's different programs over time, segmented out by categories such as member vs. non-member attendance and family vs. school attendance. We recreate a new program type chart for easier viewability, and make a visualization comparing projected signups and actual attendance. The attendance dashboard will help program managers understand which programs are happening on a given day and which have been the most popular.

- The second dashboard is **the membership dashboard**, which displays visualizations related to the museum's members across the three membership tiers. We build on the existing dashboard by adding demographic insights based on a member's city and membership sign-ups in the museum vs. online. This dashboard will help museum staff understand who they can better target to increase membership.

- The last dashboard is a **donor dashboard** analyzing donor behavior to understand giving patterns, donor retention, and engagement levels to help the museum improve fundraising efforts and growth. All these dashboards provide value and help museum staff better understand attendance, membership, and donor trends, identify outreach opportunities, and make informed strategic decisions.

1. Attendance Dashboard (UMSI Attendance Dashboard)

Developed 23 Visualization

- Track attendance trends (Compares year-over-year attendance)
- Evaluate program success
- Includes 4-week rolling averages
- Breaks down attendance by programs

Visualization	Type	Description	Purpose
Cumulative Museum Attendance Vs. 52 Weeks Ago	Line Chart	This chart shows how the attendance changes compared to the previous year.	-Track growth over time -Identify trends and patterns -Compare performance across different periods -Support decision making
Cumulative Museum Attendance Difference Vs. 52 Weeks Ago	Line Chart	This chart shows the difference in attendance between the current period and the same period 52 weeks ago.	-Measure performance compared to last year -Identify periods of growth or decline -Detect when attendance is higher or lower than previous year.

			-Support data-driven decision for programs and operations
Family Vs. School Attendance V2	Bar Chart	This chart compares attendance between two groups over time.	-Compare attendance between family visitors and school groups -Identify which group contributions more to total attendance -Track attendance patterns over time -Support planning for programs. events.
4 Week Rolling	Line Chart	This chart shows attendance trends using 4 week rolling average to smooth fluctuations	-Support planning for staffing and events -Identify patterns in attendance over time
Yankee Air Museum	Line Chart	Shows cumulative attendance over time for the Yankee Air Museum	-Identify where attendance is ahead or behind previous year. -Support strategic planning and performance evaluation.
Yankee Air Museum Cumulative Attendance Difference Vs. 52 Weeks Ago	Line Chart	Shows the difference in cumulative attendance between the current year and the same period 52 weeks ago.	-Measure overall performance compared to last year -Track whether attendance is improving or declining over time.
Birthdays Cumulative Instances Vs. 52 Weeks Ago	Line Chart	Shows the total number of birthday party events over time	Track growth in birthday bookings over time -Identify trends in event demand
Birthdays Cumulative Attendance Difference Vs. 52 Weeks Ago	Line Chart	Shows the difference in attendance for birthday parties compared to the same period last year	-Measure attendance performance vs last year -Identify the period of growth.
Birthdays Cumulative Attendees Vs. 52 Weeks Ago	Line Chart	Shows total number of birthday party attendees over time.	Track growth in birthday attendance over time -Compare current performance vs last year.

Field Trips	Line chart	This chart shows the total number of field trip attendees over time.	-Compare current performance vs last year -Identify trends in school group participation
Field Trips Instances Cumulative VS 52 Weeks Ago	Line chart	This chart shows the total number of field trip events over time.	-Track growth in field trip bookings -support planning for education programs and staffing
Field Trip Cumulative Attendance Difference Vs. 52 Weeks Ago	Line chart	This chart shows the difference in cumulative field trip attendance compared to the same period 52 weeks ago.	-Identify periods of grow or decline in field trip attendance -Mesure attendance performance vs last year.
Rentals Cumulative Attendance Vs. 52 Weeks Ago	Line chart	Shows the total number of attendees for rentals over time.	-Identify trends in event and rental usage -Track growth in rental attendance
Rentals Cumulative Attendance Difference Vs. 52 Weeks Ago	Line chart	This chart shows the difference in cumulative rental attendance compared to the same period 52 weeks ago	-Mesure rental attendance performance vs last year -Identify periods of growth or decline in rentals
Rentals Cumulative Instances Vs. 52 Weeks Ago	Line chart	This chart shows the total number of rental events (instances) over time.	-Track growth in rental booking -Compare current performance vs last year -identify trends in demand and space usage
Public Programs Cumulative Attendance vs. 52 Weeks Ago	Line chart	This chart shows the total number of attendance for public programs over time	-Track growth in public program attendance -Compare current performance vs last year -Support planning and outreach
Public Programs Cumulative Instances vs. 52 Weeks Ago	Line chart	This chart shows the difference in cumulative attendance for public programs compared to the same period 52 weeks ago	Measure public program performances vs last year -Identify periods of decline or recovery in attendance

# of Field Trips by School	Pivot table	This table displays the number of field trips (Total monthly visits) organized by each school.	-Provide a detailed breakdown of field trip activity by school and month -Identify which schools are more active - Track seasonal patterns in field trip scheduling -Support operational planning and school engagement strategies
# of Field Trip by Month (2024-Now)	Bar chart (time series)	This chart shows the number of field trips per month.	-Track monthly trends in field trip activity -Identify peak and low seasons for visits -Support planning for staffing, scheduling, and ressources -Help improve school outreach during low-activity months.
# of Field Trips by school (2024-Now)	Table	This table displays the number of field trips by school (organization)	-Identify which schools visit the museum most frequency -Compare engagement levels across schools -Help prioritize schools for future partnerships or programs
Family Vs. School Attendance (Quarter)	Bar chart	This chart compares attendance from families vs schools over time (by quarter or weekly within quarters)	-Compare attendance patterns between families and school groups -Identify which audience drives more attendances over time -Understand seasonal trend in visitors types
Family Vs. School Attendance V2	Bar chart	This chart compares family attendance vs. school attendance over time.	-Track how different audience groups contribute to total attendance - Identify which group dominates at different times

2. Membership Dashboard (UMSI Membership Dashboard)

Developed 2 Visualization

- Analyzes member vs non-member visits
- Shows signup trends (online vs in-museum)
- Provides demographic insights

Visualization	Type	Description	Purpose
Memberships Month To Date	Bar chart (Grouped bar chart)	This chart compares the number of membership sold month-to-date for the current year and the previous years. The date is grouped by sales channels	-Track current performance vs last year -compare performances across sales -Support decision making for marketing and sales
Museum Attendance vs Last Year	Line chart (Time Series Comparison)	Shows museum attendance over time using two lines	-Compare current attendance trends with last year -Identify seasonal patterns (busy vs slow periods) -Detect growth or decline over time -Support planning for staffing, events, and operations

3. Donor Insights Dashboard (UMSI Donor Insights Dashboard)

Developed 9 visualizations

- Tracks donor contributions and behavior
- Identifies retention patterns
- Supports fundraising strategies and planning event

Visualization	Type	Description	Purpose
Top 100 Donors Most Recent Donation	Scatter Plot	This chart displays individual donors as points based on the number of days since their last donation -Each dot represent a donor	-Identify active vs inactive donors -Detect donors who may need re-engagement -Understand patterns and behavior over time -Support target fundraising

			strategies
Donor Total Donated 5K - 10K	Scatter Plot	The chart displays donors based on the number of days since their last gift.	-Identify recent vs inactive donors -Understand overall donor engagement
Donor Total Donated 10K - 50K	Scatter Plot	This chart shows donors who contributed between $10,000 and $50,000	-Analize behavior of high-value donors -Identify loyal vs at-risk donors -Support target fundraising strategies
Members that Donate Breakdown	Donut Chart	This chart shows the proportion of members who donate versus those who do not	-Understand the share of members who also donate -Identify opportunities to convert members into donors -Support strategies for membership engagement and fundraising
Total Revenue by Month (2023-2025)	Bar Chart (Time series comparison)	This chart shows total revenue generated each month across multiple years (2023-2025)	-Track revenue trends over time -Compare monthly performance across years -Identify seasonal patterns in revenue -financial planning
Total Donation Per Month By Organization	Bar Chart	This chart displays the total donation amount per month, comparing two fiscal years.	-Compare monthly donation trends year-over-year - Identify peak donation periods -Track grow or decline in fundraising performance
Total Donations Per Month by Household	Bar Chart	This chart shows the total donation amount per month by households, comparing two fiscal years.	-Compare household donation trends over time -Identifywhich months households give the most -Help inform targeted fundraising strategies for households
Total # of Donations Per Month by Org	Bar Chart	This chart shows the number of donations per month, grouped by organization and compared across two fiscal years.	- Track donation frequency over time - Compare year-over-year changes in number of donations -Identify high-activity months

			for donor engagement
Total # of Donations Per Month by Household	Bar Chart	This chart shows the number of donations made by households per month, comparing two fiscal years.	- Track how often households donate over time - Compare year-over-year donation frequency -Identify month with high or low donor activity.

4. Documentation & Training Guide

- Step-by-step guide for using dashboards
- Instructions for updating data
- Best practices for long-term use

5. Tools & Technologies

- **Apache Superset** – Dashboard creation and visualization
- **SQL** – Data extraction and transformation
- **Census API** – Enriching demographic data
- **Versai**

5. Data Sources

- Museum attendance records
- Membership data
- Donor data
- External demographic data (Census API)

6. Methodology

1. Data collection
2. Data validation and consistency checks
3. Exploratory data analysis
4. Dashboard design and development
5. Client feedback and iteration

7. Key Insights (Examples)

- Attendance varies by season and program type
- Membership engagement trends differ by signup method

- Donor retention can be improved through targeted outreach

8. Challenges

- Learning new tools
  - ☐ Superset,
  - ☐ SQL environment)
  - ☐ Versai
  - ☐ Solution: Reading documentation, asking questions to our client during meetings, watching tutorial videos, Superset AI
- Navigating analysis across multiple platforms
  - ☐ Difficult to maintain consistency, keep analysis standard and organized
  - ☐ Solution: Use Gantt chart as a centralized tracker to organize tasks, monitor progress and ensure alignment on timelines / responsibilities
- Balancing technical work with client needs
- Managing project scope and deadlines

9. Impact

This project helps the museum:

- Make **data-driven decisions**
- Improve **visitor engagement**
- Strengthen **fundraising strategies**
- Better serve **underserved communities**
- Increase **sales**

10. Accessing the Dashboard

1. Active WireGuard on your computer using your username, account set up is required first by the Administrator)
2. Open your browser
3. Go to the Apache Superset link provided by the administrator
4. Log in using your credentials
5. select Dashboard in the menu bar
6. Explore visualizations
7. Use insights for planning and strategy

11. Future Work

- Automate data updates

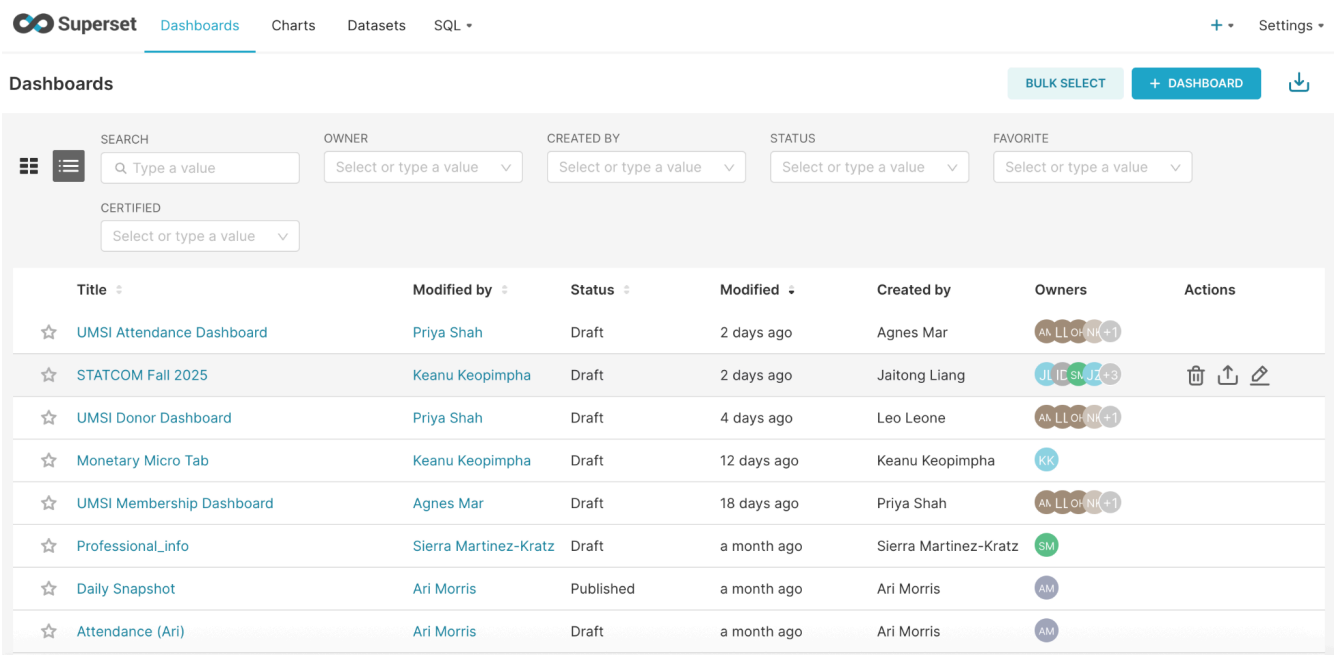
- Add predictive analytics
- Expand demographic analysis
- Improve dashboard usability

12. Team

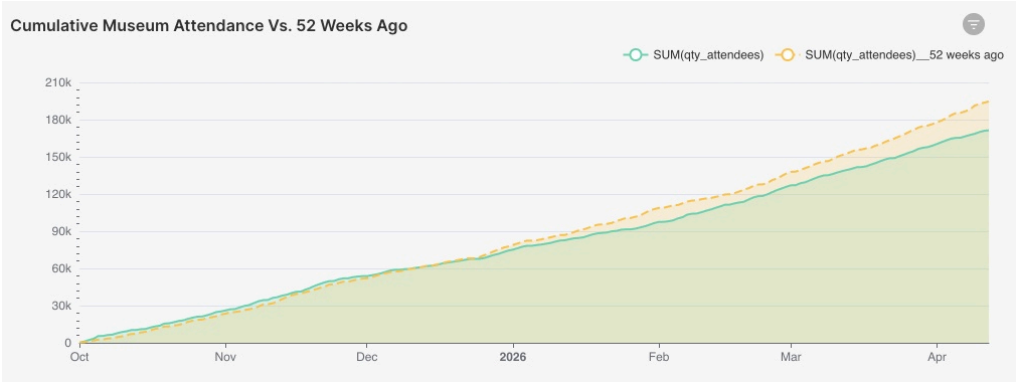
Capstone Team – University of Michigan School of Information (SI 485)

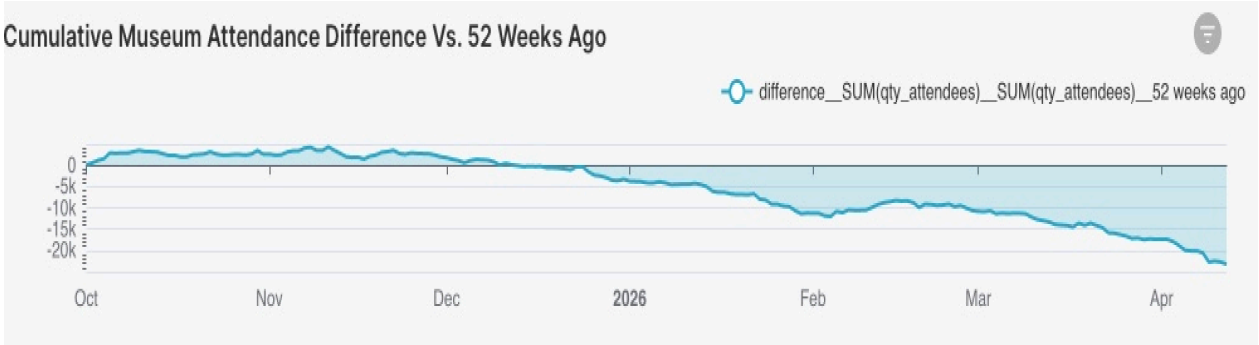
Tab 2

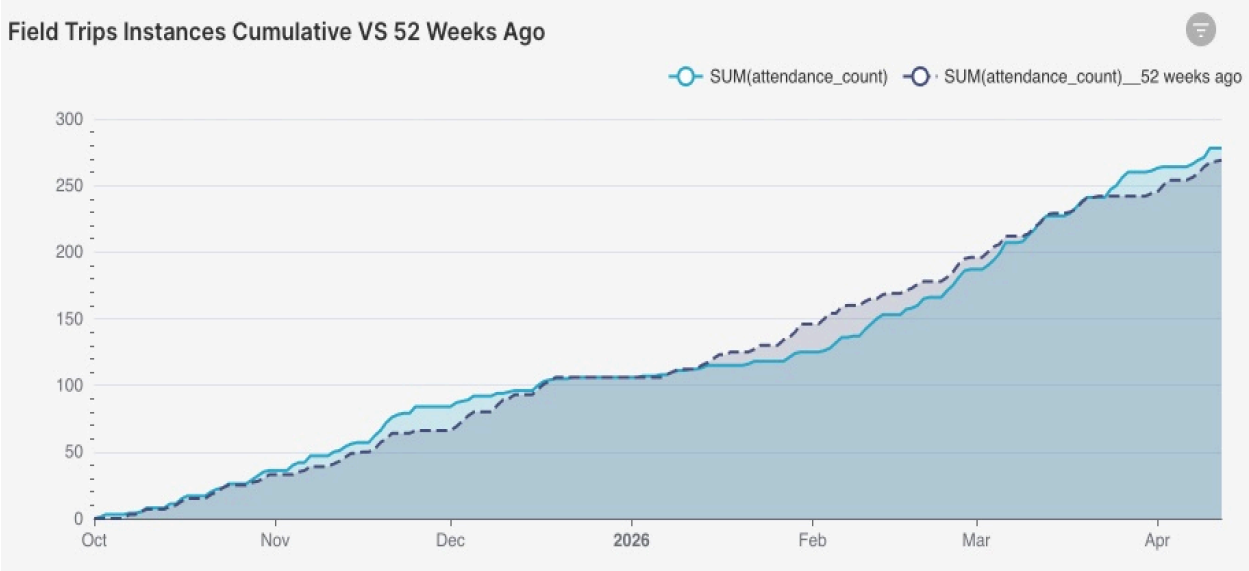
Dashboard & Visualization Gallery



1. UMSI Attendance Dashboard



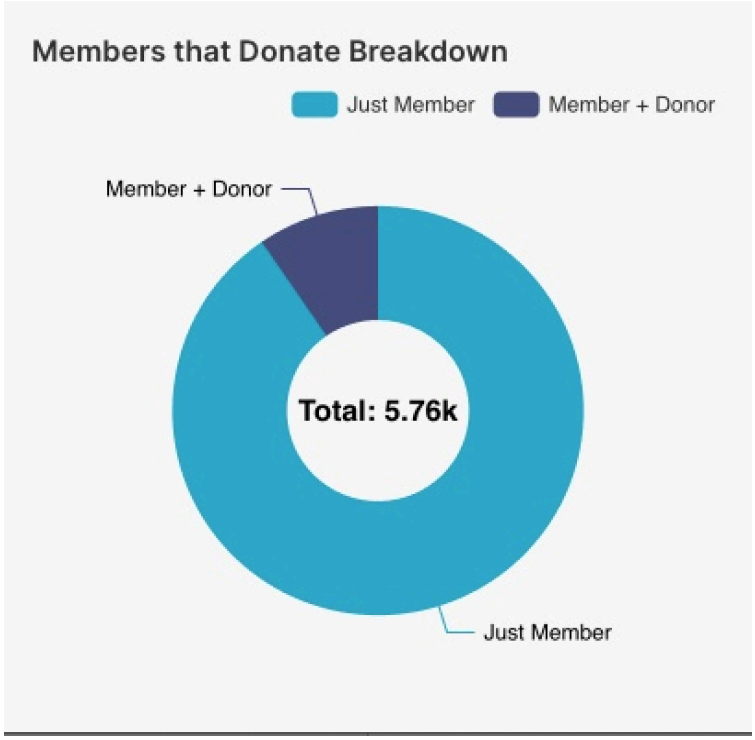


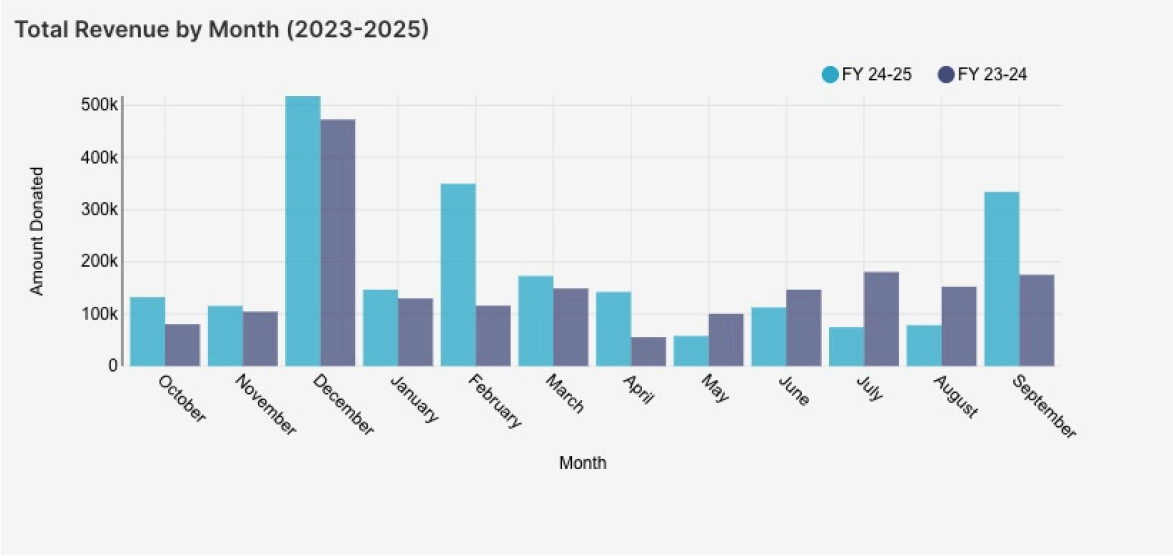

Field Trips Instances Cumulative VS 52 Weeks Ago
SUM(attendance_count)
SUM(attendance_count)__52 weeks ago
300
250
200
150
100
50
0
Oct
Nov
Dec
2026
Feb
Mar
Apr

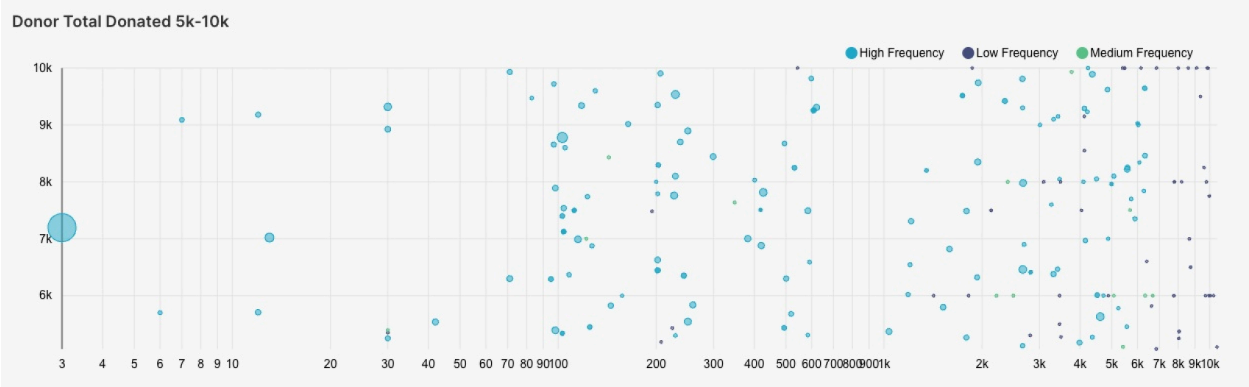
Field Trip Cumulative Attendance Difference Vs. 52 Weeks Ago
difference__SUM(qty_attendees)__SUM(qty_attendees)__52 weeks ago
4k
3k
2k
1k
0
-1k
-2k
-3k
Nov
Dec
2026
Feb
Mar
Apr

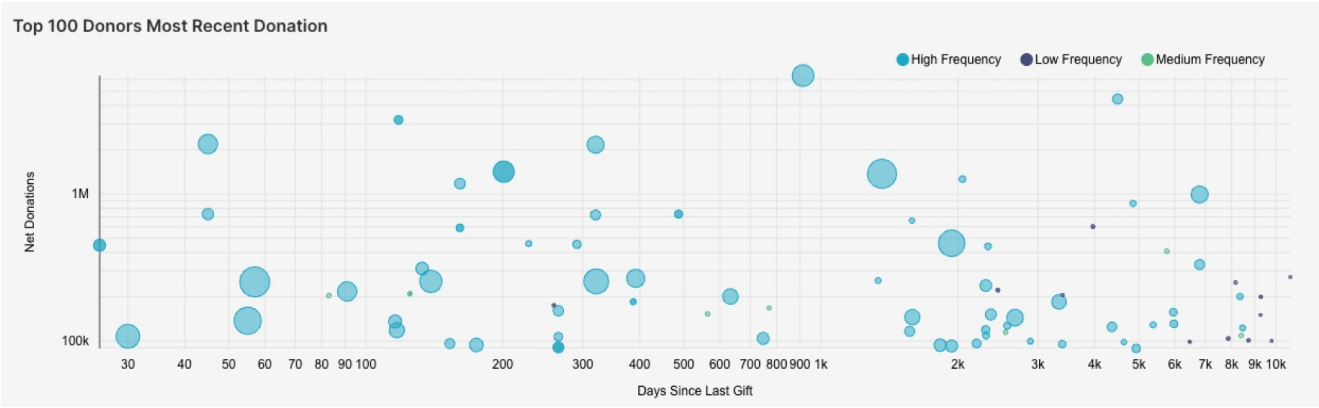
2. UMSI Membership Dashboard

3. UMSI Donor Insights Dashboard






Donor Total Donated 5k-10k
High Frequency
Low Frequency
Medium Frequency
10k
9k
8k
7k
6k
3 4 5 6 7 8 9 10 20 30 40 50 60 70 80 90 100 200 300 400 500 600 700 800 900 1k 2k 3k 4k 5k 6k 7k 8k 9k 10k


Top 100 Donors Most Recent Donation
High Frequency
Low Frequency
Medium Frequency
Net Donations
1M
100k
30 40 50 60 70 80 90 100 200 300 400 500 600 700 800 900 1k 2k 3k 4k 5k 6k 7k 8k 9k 10k
Days Since Last Gift

4. Team at the Ann Arbor Hands-On Museum

Session at AAHM